

Review retrospective:
*the harmonic regularity instance — three dead opens, a dead
import, two doc fixes*

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Abstract

A soundness review of `Laplace/OneD/HarmonicGibbsRegularity.lean`: the `Threepoint.GibbsRegularity` witness for the harmonic Gibbs measure (the harmonic analogue of the anharmonic regularity file, and the witness the harmonic cross-susceptibility capstone consumes). Result: **a clean fidelity pass with six edits** — three dead opens, a dead import, and two docstring corrections.

1 Setting

The harmonic counterpart of `AnharmonicGibbsRegularity`: it proves the regularity data for $L = (\lambda/2)x^2$, $A = \text{id}$ via the *closed form* of the perturbed partition (square completion + Lebesgue translation + `integral_gaussian`) rather than dominated convergence, and is the substrate the harmonic cross-susc capstone builds on.

2 What was reviewed

The perturbed partition closed form $\int e^{-t((\lambda/2)x^2 + hx)} = e^{th^2/2\lambda} \sqrt{2\pi/\lambda t}$, its $h = 0$ specialisation, $Z > 0$, the first-moment odd-symmetry vanishing $\int (-tx)e^{-(t\lambda/2)x^2} = 0$, and the `harmonic_id_gibbsRegularity` assembly.

3 Fidelity / soundness findings

Sound; no defects. Square completion $(\lambda/2)x^2 + hx = (\lambda/2)(x + h/\lambda)^2 - h^2/2\lambda$ gives the closed form via translation invariance and `integral_gaussian` at $b = t\lambda/2$; $Z > 0$ via `Real.sqrt_pos`; the first moment vanishes by odd symmetry ($x \mapsto -x$ invariance); and the three `GibbsRegularity` fields assemble correctly. The reviewer confirmed all four (noting, harmlessly, that the odd-vanishing lemma is stated for all λ, t because Lean totalises the Bochner integral — true as written, used under $\lambda, t > 0$).

Two docstring corrections. The file header wrote the potential as “ $((\lambda/2)\cdot)$ ” — linear scaling, not the quadratic $(\lambda/2)x^2$ (the *headline* bullet already had it right); corrected to match. And the helpers bullet advertised `harmonic_integral_x_exp_neg_eq_zero` (wrong name, and its formula dropped the $-t$), while the actual theorem is `harmonic_integral_neg_t_x_exp_neg_eq_zero`; corrected name and formula.

4 Simplifications applied

Three dead opens and a dead import. `open Real` (all `Real.*` qualified, bare $\sqrt{\pi}$ only in docstrings), `open Filter` (all `Filter.*` qualified, no `filter_upwards`), and `open Topology` (uses root-level `nhds`, never $\mathcal{N}$) are all dead; `open MeasureTheory` stays. And `import ...IntervalIntegral.Basic` is dead — no interval-integral notation or lemmas are used (`integral_gaussian` comes from `GaussianIntegral`); a full build confirmed it is not transitively load-bearing. Signatures byte-identical, no warnings. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s proof-golf (`show`→`change`, a `simp`-collapse of `harmonic_partition_eq`) was declined.

6 Lessons

- **The headline bullet and the file header can disagree — the header was the wrong one.** The `## Headline` section already wrote the potential as a `fun x => ...` mapping $x \mapsto (\lambda/2)x^2$; only the top-of-file summary had the lossy “ $((\lambda/2)\cdot)$ ”. When two docstring sites state the same object, cross-check them against each other (and the actual signature), not just against intuition.
- **A four-open header in a closed-form-Gaussian file is mostly dead.** Three of the four opens (`Real`, `Filter`, `Topology`) carried nothing — the bare-name greps called each before the build confirmed. Headers accrete opens from the template; only `MeasureTheory` (the bare $\int$) was load-bearing.

7 Follow-ups

- The remaining harmonic files: `HarmonicGibbsObservableMonomials`, `HarmonicAffineKappa3`, `HarmonicPerturbedMoments`; plus `QuarticBoundedPrior/QuarticBoundedPriorTestFn` and `IntegralRemainder`.